

Machine Learning and DATA

DATA:

The Collection of useful information.

Type of Data:

1. **Structured DATA:** Schema is Defined. Example: Tabular data
2. **Unstructured DATA:** Schema is Undefined. Example: Image, audio
3. **SemiStructured:** SemiStructured data is data that does not reside in a traditional relational database but still has some organizational properties, such as tags or markers that make it easier to analyze than completely unstructured data.

How data is stored?

Different type of data stored in different format

Structured Data stored in the Tabular format.

1. CSV (Comma-Separated Values)
2. `.xlsx` (Excel)

Unstructured Data and Semi-Structured Data stored in

1. JSON (JavaScript Object Notation)
2. XML (eXtensible Markup Language)
3. image

Labeled data: contains tags (like "cat" or "spam") that define the target answer.

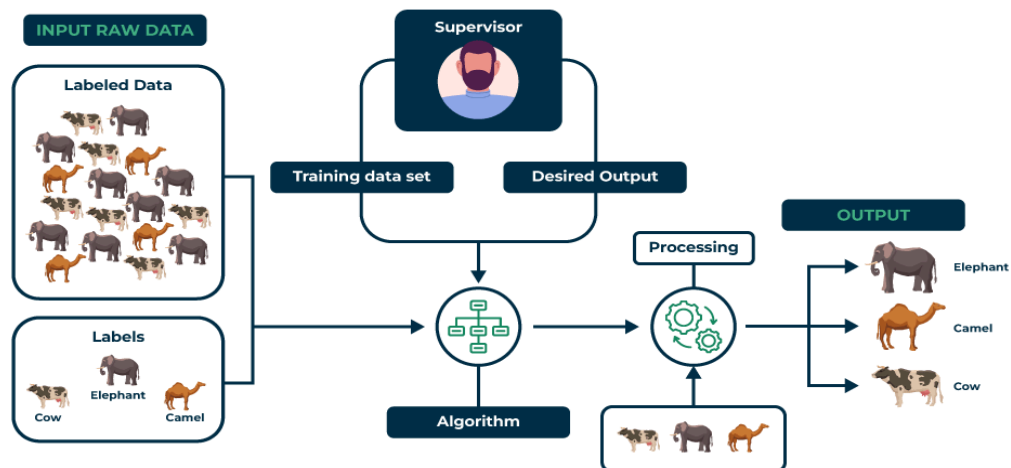
Unlabeled data: raw or untagged information

MACHINE LEARNING:

Machine Learning Teaches the machine to learn from the data , detecting the relations of the data.

Type of machine learning based on the data:

1. **Supervised ML:** The model which works on the labeled data.



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2. **Unsupervised ML:** The model which works on the unlabeled data.

